

A MODEL AND AN ANALYTICAL METHOD FOR CONVEYOR SYSTEMS IN DISTRIBUTION CENTERS

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Abstract

The design and operation of high volume conveyor systems in distribution centers are important due to its high cost, large footprint and critical role in the system. However, there is no analytical model available. In this paper, we study the characteristics of the conveyor sortation system, develop a generic model of a complex conveyor network. We then use the Delay and Stock abstraction to develop an approximate analytical method using sample path analysis and dynamic network flow model. The decision variables include length of accumulation segments and the speeds of conveyor components aforementioned decision variables. This analytical solution provides faster analysis, insights and useful subgradient with respect to the decision variables.

Keywords: Conveyor system, sortation system, network, accumulation, sample path analysis, dynamic network flow

List of Notations

Notation on conveyor system

$sm = (d, sp, l, c, IN, OUT)$	A conveyor segment in the set SM . A segment is represented as an arc $e \in A$ in the network
d	Conveyor segment maximum density, items/unit length
sp	Conveyor segment driving speed, meter/unit time
l	Conveyor segment length
$c \in \{ 'A', 'N' \}$	Conveyor segment type, 'A' for accumulation and 'N' for non-accumulation
IN	A set of junctions that link with sm at ENTER end
OUT	A set of junctions that link with sm at EXIT end
τ	Conveyor segment traverse time, $\tau = \frac{l}{sp}$

$jn = (r, b, \sigma, IN, OUT)$	A conveyor junction in the set JN , a junction is modeled as a node $v \in V$
r	Maximum transfer capacity of a junction, items/unit time, ∞ for pure connection
b	Maximum buffer capacity of a junction, 0 for pure connection
F	Feasible set of parameters for l, b, sp
σ	Transfer control logic of a junction (based on buffer occupation rate, priority merge/diverge, priority merge/diverge, round-robin merge/diverge, equal time merge/diverge, etc.). It is also used as a parameter in iteration, clear in the contexts.
IN	A set of upstream conveyor segments and junctions that link with jn
OUT	A set of downstream conveyor segments and junctions that link with jn
$CN = (SM, JN, SIN, s)$	A conveyor network
$SIN \subset JN$	The subset of junctions that receive external input
m	The cardinality of subset SIN
$s \in SM \cup JN$	The sink with $OUT(s) = \emptyset$

Notation used to model the conveyor as networks

$\mathcal{N} = (V, A)$	Network representation of the conveyor
$v \in V$	Node set. Each node represents a junction
$\delta^+(v), \delta^-(v)$	Sets of leaving or arriving nodes
$e \in A$	Arc set. Each arc represents a conveyor segment
u_e	Arc capacity
z_v	Node capacity
τ_e	Transit time in an arc
Δ	Base time unit in discretization
p	Number of base time units for the time horizon
$m_e = \frac{\tau_e}{\Delta}$	Integer fraction of transit time on a conveyor segment

Notation on flows

$\Lambda(t) = (\Lambda_1(t), \dots, \Lambda_m(t))$	Arrival process from different sources, assuming $\Lambda_i(t)$ and $\Lambda_j(t)$ are independent.
$\lambda = (\lambda_1(t), \dots, \lambda_m(t))$	Arrival rate immediately after time t at junction $i \in SIN$.
$\mathcal{J}(t)$	Input flow rate to the ENTER side of the segment, cartons/minute
$\mathcal{O}(t)$	Discharge capacity at the EXIT end of the segment; this is the maximum rate at which items can be removed from this segment
$I(t)$	Effective in-flow rate or the achieved rate at which items will be loaded to this segment;
$O(t)$	Effective out-flow rate; this is the real achieved rate at which items will be unloaded from this segment

$a(t)$	Accumulation position.
$\mathcal{L}, L$	Cumulative loss volume computed from random input process $\Lambda(t)$ and sample path $\lambda(t)$.
H	A percentage item loss predetermined for the design problem.
g	The expected loss in time horizon T
Q	Total flow volume to the sink
Parameters used in solutions	
θ	Set of decision variables
r_k	Parameter for penalty weight
σ	A non-normalization vector to adjust the big difference of magnitude of the two parameters in iteration. It is also used as transfer control logic. They are clear in the context.

1. Introduction

To support increasing throughput in the fierce world of retailing, logistics service providers and retailers are consolidating their distribution business by expanding or building large distribution centers (DCs). These DCs support large customer base and high order fulfillment capacity. The daily throughput can be hundreds of thousands of items. Conveyors and high speed sortation systems become desirable when the throughput reaches a certain level. In our survey of 13 retailers of general merchandise, toys, drugs, books, cosmetics, apparel, the average size of the DCs is 67,781 m², daily throughput 163,333 items, 428 clients (excluding two direct shipping facilities), and 13 miles of conveyors. The conveyors in these systems expand the pick and loading area throughout the facility to allow higher labor efficiency and higher throughput. The items are loaded and picked to the conveyor, merged again and again to form a high speed stream of items for final sortation to the shipping lanes. These conveyor sortation systems are characterized by large footprint, complex configurations,

high-throughput, high WIP and high cost. The design and control of these systems have significant impact on the system performance such as throughput, order response time and operating cost.

High throughput DCs have tens of thousands of SKUs and many of them are picked frequently. The required pick area is normally very large. To improve the pick rate and keep the pick tours short, a large pick area is often divided into pick zones. Conveyors extend any area into a good pick area by connecting it to the sorter. The inducting, transporting, accumulating, merging and sorting are the main functions of a conveyor sortation system. We can decompose a sortation system into three main subsystems: (1) Induction, including order picking and receiving, (2) Accumulation and Merge, and (3) Sort and Post Sort, conceptually partitioned and shown in Figure 1.

Conveyor induction at receiving is for flow through or cross dock items. The order picking consists of one or more picking zones. Pickers at picking zones pick to the conveyor, which transports, merges and routes the items toward

the sorter. The challenge in conveyor system is to synchronize the flow from the various picking zones (Apple 1999) because of varying travel distances, pick rates and item density on the conveyor.

Accumulation and merge subsystem is the

conveyor system that collects items from picking and receiving, accumulates, merges and remerges, and prepares for sortation. Its objective is to provide a uniform and high density flow of items to the sortation system. This high density, uniform stream is achieved

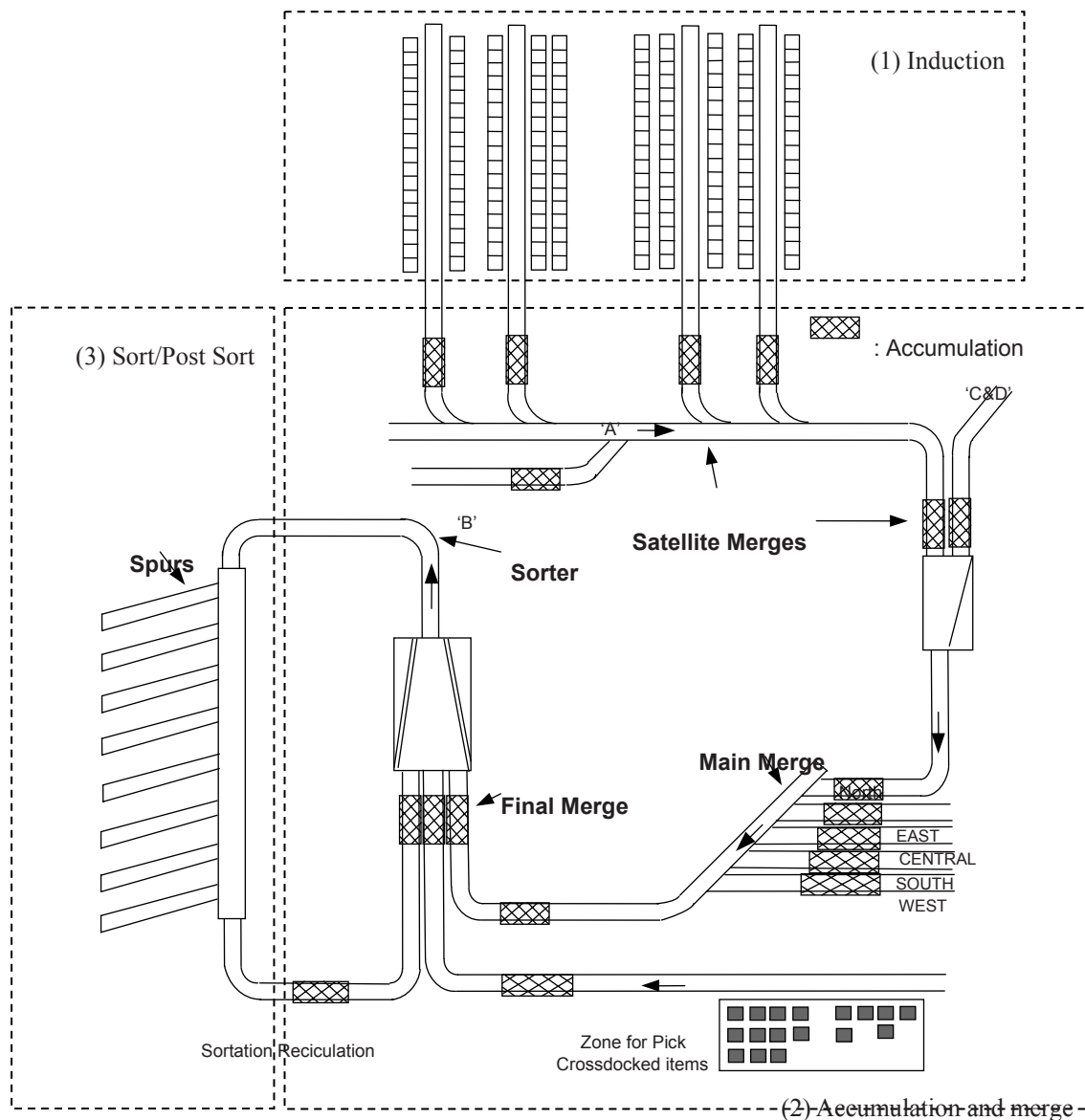


Figure 1 Subsystems of conveyor sortation system

through accumulation, buffering at junctions, speed control and merge logic. The accumulation and buffers are used to buffer against the peaks and valleys of the instantaneous input rates variations from different sources. The accumulation conveyor and buffer at junctions incur significant cost. It is important during the design stage to select proper accumulation lengths, speeds of conveyors and junctions, and the buffers at junctions (Maxwell & Wilson 1981). The decisions related to the accumulation and merge subsystem design include:

- Conveyor sortation system topology
- Lengths of the accumulating and non-accumulating conveyor segments
- Transfer rate and buffering at junctions
- Merge and diverge control policy design

Sortation system diverges streams of items into takeaway conveyors that link to the docks for different destinations. The takeaway conveyor provides buffering to reduce the probability of blocking. Sort and Post Sort systems may be classified (Bozer 1988) into two types: S/PS-1 and S/PS-2. In S/PS-1, where the number of lanes is less than the number of orders; such as e-tailers (e.g. Amazon.com) and catalog distributors (e.g. J.Crew). The items without lane assignment are re-circulated. The major bottleneck of the S/PS-1 process is normally the recirculation loop. In S/PS-2, the number of lanes is equal to or more than the number of orders. Lanes are provided so that every incoming item will be destined to a particular lane during a wave. Recirculation occurs only when the corresponding lane is full or identification fails. This research will focus on the S/PS-2.

There is a variety of research related to conveyors and sortation systems. We can summarize our literature review results into following observations:

1) Conveyor systems in large DCs are expensive. The construction cost of an automated DC varies in the 10-50 million dollars range. The hourly operating expenses often exceed 1,000 dollars (Johnson 1998).

2) Past research on sortation systems focus on type 1 Sort/Postort subsystem design (Johnson 1998, Johnson & Russel 2002, Russel & Meller 2003). Research on accumulation /merge subsystem design is limited to simulation study or aggregated analysis in which the subsystem was treated as a large static buffer (Bozer & Sharp 1985, Bozer et al. 1988). However, the large footprint, complex topology and length of high cost cumulating conveyor segments are important in the efficiency of the design.

3) There are papers on general conveyor theory, but they mainly focus on the production system served by a close-loop system with complex control but simple topologies (Kwo 1958, Bastani 1988).

4) Stochastic conveyor network analysis has been restricted to very simple structure with one or two junction nodes (Zrnic & Cupric 1992).

5) In large scale conveyor systems, the item-level simulation can be too slow for design and optional control. We had developed a Batches Petri Net and fluid based simulation analysis (Wang & Zhou 2004, 2005).

The Batches Petri Net and fluid based simulation is faster than item based discrete event simulation. However, the simulation based analysis is still slow. This research is an attempt

to understand the dynamics and to develop an analytical method for rough cut analysis. The Delay and Stock abstraction approximates the spatial material flow evolution. However, there has been no study done to evaluate the accuracy of this representation.

1.1 Objectives

We will focus on the accumulation and merge subsystem in a type 2 sortation system where the number of lanes is equal to or greater than the number of orders. In the four sets of decision variables, our objective is to find the length of high cost accumulative segments in a conveyor system, the transfer rate and the buffering at the junctions. The model can be used for feasibility study in the design stage. We integrate the stochastic optimization techniques with deterministic system evaluation tools to search for the near optimal design with minimum cost and performance bounds.

1.2 Conveyor System Generic Model

Accumulation and merge conveyor network shown in Figure 1 can be modeled as a network

shown in Figure 2.

The basic elements of this conveyor network abstraction are conveyor segments and junctions. A conveyor segment is a one-directional transportation and buffering device, which receives items from one end and sends them to another end preserving the sequence of items during transportation. We call the two ends ENTER and EXIT respectively. Mathematically, we represent a segment by the structure $sm = \{d, sp, l, c, IN, OUT\}$. We use junction to model the pure connection between two segments, a merge/diverge control point or a station with or without a buffer. We also define a virtual ENTER side and EXIT side for junction even though there is no physical distance between ENTER and EXIT of a junction. All upstream elements of the junction connect to the ENTER side while the EXIT side connects with all downstream elements. It transfers items from upstream elements to downstream elements according to some transfer control logic. We represent a junction formally by $jn = \{r, b, \sigma, IN, OUT\}$. The connection among elements is defined through IN and OUT

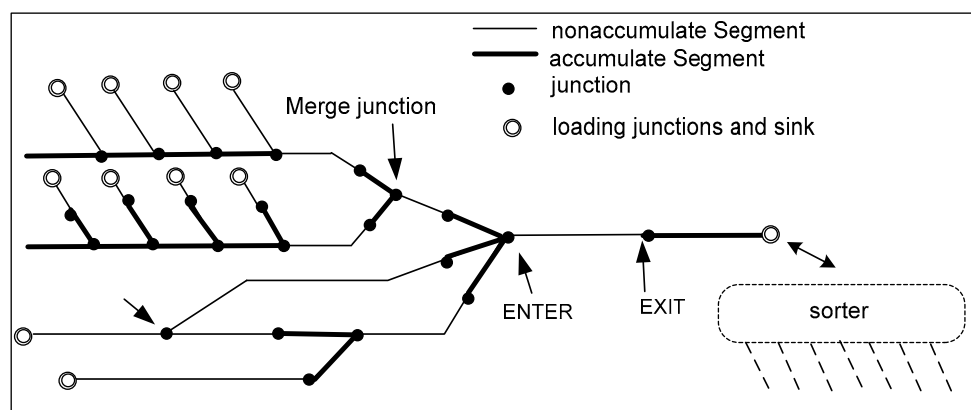


Figure 2 Accumulation and merge conveyor network

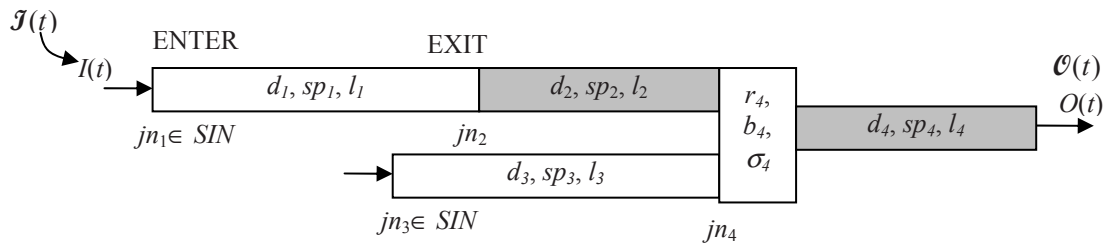


Figure 3 Conveyor system model schematic

structure. That means if an element i belongs to the IN set of element j , i.e. $i \in IN_j$, then the ENTER of element j overlaps with the EXIT of i .

Using segment and junction, an accumulation and merge conveyor network can be defined by the structure $CN = (SM, JN, SIN, s)$. External arrivals can enter the ENTER end of a segment. In this case, we assign a virtual junction for modeling purposes. Similarly, if the external arrival is at some middle point of a conveyor segment, the segment will be split into two in the model with a junction inserted in between.

1.3 Assumptions

1. The general topology and layout of the conveyor network is given. The focus of the model is to find the length of accumulation conveyor segments, the transfer rate and buffer of the junctions. The conveyor driving speed is fixed for each evaluation. Different speeds can be evaluated in successive analysis.

2. Uniform product length. This is a good approximation when the products are of similar sizes, such as in case picking, or piece picking using uniform shipping cartons.

3. Sortation subsystem is of type 2, or the number of lanes in the sorter is greater than or

equal to the number of shipment destinations. We further assume that the lane capacity is sufficient, and recirculation is omitted.

1.4 Arrival Process

We consider arrivals over a continuous time horizon $[0, T]$. T can be a shift, a day or a wave. The arrival process is stochastic because of the random variations in orders, the order picking process and the interruptions. In discrete modeling, Poisson arrival is commonly used. Closer observation reveals that items tend to arrive in chunks of different densities. The loading of cases at receiving is idle for most of the time, but has a steady rate of loading for some periods. This bursty nature does not agree with Poisson arrival assumptions. Also, the variations take place in different time scales. The time for item interarrivals are in seconds, while the time for rate change is usually in minutes or tens of minutes. The queuing effect contributed from variability of larger time scale is significant and dominant compared to small time scale effect. We need a stochastic process that can better capture this larger time scale variability.

We adopt a stochastic fluid model in describing the arrival process. Let $\Lambda_i(t), i = 1, \dots, m; t \in [0, T]$ be a stochastic process taking values in $\mathbb{R}^+$ and

$\lambda_i(t), i = 1, \dots, m; t \in [0, T]$ a sample point of $\Lambda_i(t)$ from its sample space. $\lambda_i(t)$ represents the arrival rate immediately after time t at junction $i \in SIN$. We also assume $\Lambda_i(t)$ has a piece-wise constant sample path. We use $\Lambda = (\Lambda_1(t), \dots, \Lambda_m(t))$ and $\lambda = (\lambda_1(t), \dots, \lambda_m(t))$ to denote the vector form of random input process and its sample path. We assume that the arrival rate of two sources $\Lambda_i(t)$ and $\Lambda_j(t)$ are independent. Although fitting a stochastic input model Λ for a given system is itself a challenging problem, we will assume that a good fitting and generating mechanism has already been developed so that we can focus on the network optimization.

1.5 Costs of Conveyor Components

The system cost considered in the model consists of fixed and variable costs associated with conveyor segments, merge devices and sorter.

Conveyor segment fixed cost depends on equipment type and accumulation type. For example, a powered live roller conveyor ranges from \$200 – \$250/ft; a powered slider bed conveyor ranges from \$180 – \$250/ft; while a powered accumulation conveyor ranges from \$250 – \$350/ft. In our model, we assume that the fixed cost of the segment only depends on accumulation type c and speed sp . The variable cost increases with speed.

The increase in merge device sophistication and rates r , measured by items per minute, directly leads to a higher fixed cost and variable cost. For example, live roller merge and herringbone merge are considered low-to-medium rate merge in the range of 10 – 80 cartons/min; while sliding shoe merge and

saw-tooth merge are capable of rates in excess of 80 cartons/min.

Sorters can be classified by three rate categories: low, medium and high, or 0 – 40, 40 – 80 and 80 – 200 cartons per minute, respectively. Example low rate sorters are 90 degree transfers, barrier-type diverters and pusher sorters; example medium rate sorters are pop-up wheel and roller sorters; while sliding shoe sorters and tilt tray sorters are considered as high rate sorters. Usually, higher rate sorters are much more expensive than lower rate sorters.

1.6 Performance Measure

One of the most important design objectives for a high volume order fulfillment system is to achieve its required throughput via accumulation and merge control. High throughput requires an accumulation and merges subsystem to present a uniform and high density item stream to the sorter. The conveyor speed, the length of accumulation and merge control are important variables to buffer and stream line the imbalances of arrival and order picking processes. A good design should achieve minimum voids in the item stream to the sorter simultaneously, and minimum blocking in arrival processes and with minimum cost.

We use the loss volume $\mathcal{L}$ defined below as a performance measure to evaluate the feasibility of the design. This assumption is acceptable when the loss is small. By letting $\Psi(t)$ be the cumulative arrival process to the sink, the loss volume can be defined as

$$\mathcal{L} = \sum_{i \in [1, \dots, m]} \int_0^T \Lambda_i(t) dt - \int_0^T \Psi(t) dt \quad (1)$$

$\mathcal{L}$ is a random variable since Λ and $\Psi(t)$ are

stochastic.

1.7 The Development of the Optimization Model

The objective of the model is to determine the set of parameters of a minimum cost design that satisfies the maximum loss percentage requirement in the stochastic work environment. The decision variables in this model are accumulation conveyor segment lengths, junction capacity and buffer size, denoted as $\{l_k \mid k \in SM\} \cup \{r_j, b_j \mid j \in JN \setminus SIN\}$. The sorter speed is represented as one of the junction capacity. The number of decision variables is equal to the cardinality of the segment set plus twice of the cardinality of the junction set (excluding the junctions representing the external inputs). We let $\theta \in \mathbb{R}^D$, $D = |SM| + 2 \times (|JN \setminus SIN|)$ denote the complete set of decision variables.

The design problem can be formulated into the following optimization problem:

Problem 1: Minimize $C(\theta)$

s.t.

$$g \equiv E[\mathcal{L}(\Lambda, \theta)] \leq E \left(\sum_{i \in [1, \dots, m]} \int_0^T \Lambda_i(t) dt \right) \times H\%$$

$$\theta \in F$$

where C is some convex cost function, $\mathcal{L}(\Lambda, \theta)$ is the random variable representing the loss volume for a given θ ; g is the expected loss volume and H is pre-selected system percentage loss limit. F represents the feasible set for parameter l, r, b that satisfies the facility dimension and equipment capacity constraints.

Problem 1 is a constrained stochastic optimization problem with linear or nonlinear objective function and nonlinear stochastic constraints. There is no closed form expression

for the loss volume even for a deterministic input stream. The lost volume can only be found by simulation or approximated by another optimization problem. Various simulation optimization techniques can be used, with the choice based on the assumption of the feasible region. Our decision variables are the length of each accumulation conveyor segment and the speed of merging and sorting at each junction. If we treat these variables as discrete, the feasible region is finite but too large. If we treat decision variables as continuous, we can use some gradient-based search method such as stochastic approximation. Stochastic approximation schemes generally require convex and differentiable objective functions but the simulated performance function may not be differentiable.

Another approach is sample path optimization presented and analyzed by Gurkan (1994), and Plambeck et al. (1996). It is also called sample average approximation in Shapiro (2003). The basic idea of such methods is to approximate the expectation function through Monte Carlo sampling, the consequent sample average approximation problem becomes deterministic and the powerful machinery of mathematical optimization can be applied directly. The optimal solution of this sample average approximation problem is taken as an estimator of the “true” optimizer. Under certain conditions, this approach can be shown to converge almost surely as the sample size of Monte Carlo sampling increases. We adopt this approach in the following solution procedure because of the non-smooth nature of our problem and the desire of fast convergence.

In problem 1, we have two types of

constraints. The feasible set F is given by explicit, simple linear constraints. They represent the physical bound of the system and parameters outside these constraints are invalid. So they are considered as *hard* constraints. These constraints are better handled by projection method in numerical solution procedure. The second set of constraints imposed by loss volume performance measure is stochastic and often nonlinear, non-differentiable. This constraint is considered *soft* which means we allow parameters to take on values violating this constraint during optimization. This type of constraint can be incorporated into objective function in a form of penalty. Problem 1 will then be transformed to a sequence of mildly constrained problem:

Problem 2: Minimize

$$C(\theta_k) + r_k P \left(E[\mathcal{L}(\Lambda, \theta_k)] - E \left[\sum_{i \in [1, \dots, m]} \int_0^T \Lambda_i(t) dt \right] \times H\% \right)$$

s.t. $\theta_k \in F$

Penalty function $P(x)$ should be defined satisfying the following conditions:

$P(x)$ is differentiable on $\mathbb{R}$

$P(x) \geq 0$ for all $x \in \mathbb{R}$

$P(x) = 0$ if and only if $x \leq 0$

$P(\cdot)$ is strictly increasing on $[0, \infty)$

The r_k are a sequence of penalty weights with $r_k > 0$ and $r_k \rightarrow \infty$.

We solve the problem 2 by sample path optimization, that is, by generating random samples $\lambda^1, \lambda^2, \dots, \lambda^N$ of Λ and approximating problem 2 by its corresponding sample average problem:

Problem 3: Minimize

$$U(\theta_k) \equiv C(\theta_k) + r_k \cdot P(g_N(\theta_k) - M)$$

s.t. $\theta_k \in F$

where $g_N(\theta_k) = N^{-1} \sum_{j=1}^N L(\lambda^j, \theta_k)$ and

$$M = \left(N^{-1} \sum_{j=1}^N \sum_{i \in [1, \dots, m]} \int_0^T \lambda_i^j(t) dt \right) \times H\%. M \text{ is a}$$

constant for a fix set of samples.

We assume that C and g are convex with respect to θ . Due to the possibility that g is non-smooth, we employ projection sub-gradient optimization procedure to solve problem 3. The iterative scheme of this approach has the form:

$$\theta_{k+1} = \Pi_F(\theta_k + a_k \cdot s(U))$$

where $s(U)$ is a subgradient of U at θ_k and calculated by

$$s(U) = \nabla C(\theta_k) + r_k \cdot \nabla P(g_N(\theta_k) - M) \cdot s(g)$$

∇ is the gradient operator and $s(g)$ is a subgradient vector of $g_N(\theta_k)$.

$\{a_k\}$ is a positive sequence of numbers converging to 0, and Π_F denotes a projection operator that maps points in $\mathbb{R}^D$ to their nearest neighbor in F . In the following section, we investigate the method that evaluates $g_N(\theta_k)$ and $s(g)$.

2. Analytic Conveyor Network Performance Evaluation Model

In Wang & Zhou (2004, 2005), we developed a fluid simulation model to solve the problem. The simulation based method provides a more accurate solution to the problem since the model keeps track of conveyor state evolution in time and space. However, it is more time consuming. In this paper, we will develop an analytical method. We use the Delay and Stock abstraction to develop a deterministic optimization model. The model evaluates the loss volume for a single

replication of Monte Carlo sampling as a parametric optimization problem. The resulting model is a class of dynamic network flow problem that we call the maximum flow over time problem with piece-wise constant arc capacity and constant finite node capacity. To solve the problem, we transfer it to a linear program through uniform discretization. Not only does this model give us the loss volume, but it also provides us with a subgradient of loss volume with respect to the decision variables. This can be incorporated into our parametric optimization solution framework in Problem 3.

2.1 Delay and Stock Abstraction

In the Delay and Stock abstraction a conveyor segment is modeled by a fixed delay τ from its ENTER end to its EXIT end, and provides a fixed size buffer at the EXIT end of the conveyor. If it is a non-accumulation conveyor, the buffer size is zero. Delay and Stock abstraction expresses conveyor capacity and accumulation constraints mathematically through the following equations and inequalities:

$$\tau = \frac{l}{sp} \quad (2)$$

$$\int_0^t I(\theta - \tau) d\theta = \int_0^t O(\theta) d\theta + a(t) \quad \forall t \in [0, T] \quad (3)$$

$$I(t) \leq J(t) \quad \forall t \in [0, T] \quad (4)$$

$$O(t) \leq \mathcal{O}(t) \quad \forall t \in [0, T] \quad (5)$$

$$I(t) \leq sp \cdot d \quad \forall t \in [0, T] \quad (6)$$

$$O(t) \leq sp \cdot d \quad \forall t \in [0, T] \quad (7)$$

$$\begin{cases} a(t) = 0 & \text{if } c = 'N' \\ a(t) \leq \frac{l}{sp} & \text{if } c = 'A' \end{cases} \quad \forall t \in [0, T] \quad (8)$$

(2) states the fixed traverse time is equal to the segment length divided by its speed. (3) is the flow balance equation followed by material conservation law. (4) and (5) limit the effective in-flow and out-flow rate to the input flow rate and output capacity. (6) and (7) describes the flow rate limits imposed by transportation capacity of segment. (8) requires that the accumulation position will never exceed the maximum buffer size. At each instant, $I(t)$ and $O(t)$ attain maximum values that satisfy conditions stated in (2) to (8).

2.2 Maximum Flow over Time Model

Based on the Delay and Stock abstraction, we will construct an analytical model for performance evaluation with a given deterministic input stream. We first introduce the model and then give a transformation procedure for transforming the data in a conveyor network to the input data of the analytical network model.

A *maximum flow over time* problem with piece-wise constant arc capacity and constant node capacity is defined on a networks $\mathcal{N} = (V, A)$ with $n := |V|$ nodes and $m := |A|$ arcs. Each arc $e \in A$ has an associated integral transit time or length τ_e and a non-negative, left-continuous, piece-wise constant capacity function $u_e: (0, T] \rightarrow \mathbb{R}^+ \cup \{0\}$. We can scale time such that the break points of all u functions are integral multiple of time units. Each node $v \in V$ has an associated integral capacity z_v . Let $S^+ \subseteq V$ be a subset of sources and $S^- \subseteq V$ a single sink. The problem is to find a feasible flow over time that sends as much flow as possible from sources to the sink in time T .

A flow over time f on $\mathcal{N}$ with time horizon

T is given by a collection of Lebesgue-measurable functions $f_e: [0, T] \rightarrow \mathbb{R}^+$ where $f_e(\theta)$ determines the rate of flow (per time unit) entering arc e at time θ . The problem can be formulated as follows.

$$\begin{aligned} \max \quad & \int_0^T \sum_{e \in \delta^-(S^-)} f_e(\theta - \tau_e) d\theta \\ 0 \leq \int_0^t & \left[\sum_{e \in \delta^-(v)} f_e(\theta - \tau_e) - \sum_{e \in \delta^+(v)} f_e(\theta) \right] d\theta \leq z_v \\ & \text{for all } t \in [0, T], v \in V \setminus S^+ \end{aligned} \quad (9)$$

$$f_e(\theta) \leq u_e(\theta) \quad \text{for all } \theta \in [0, T] \text{ and } e \in A \quad (10)$$

The objective is to maximize total flow to the sink during the time horizon. (9) is the flow balance constraint which states that for any node v other than source, at any time instant t , the total inflow to a node v minus total outflow of node v must be non-negative but less than the node capacity. Here $\delta^+(v)$ and $\delta^-(v)$ denote the set of arcs leaving node v and entering node v , respectively. Constraints (10) require a feasible f obey the capacity constraints through the time horizon.

For a given accumulation and merge network and input stream, the transformation to an instance of maximum flow over time problem in three steps is presented in Appendix A. It can be verified that this transformation procedure enforced the solution of the maximum flow over time problem, and satisfies all capacity and accumulation constraints in the original network expressed through (2) to (8).

2.3 Related Model and Solutions in the Literature

Maximum flow over time model falls into the class of dynamic network flow problem

(DNFP). Dynamic network flow problems generalize standard network flow problem by introducing an element of time to model problems where travel is not instantaneous or when network capacities restrict the quantity of flow that can be sent at any one time, and thus necessitate sending flows in phases.

Within DNFP problem class, *maximum dynamic flow problem* was defined similarly as in our problem, but it has constant arc capacity and infinite node capacity. Ford & Fulkerson (1962) showed that the maximum dynamic flow problem could be solved in polynomial time via one minimum-cost flow computation. *Quickest transshipment problem* is defined in a dynamic network with a set of sources and sinks; each source has a specified supply of flow and each sink has a specified demand. The problem is to send exactly the right amount of flow out of each source and into each sink in the minimum overall time. Hoppe & Tardos (2000) give the first polynomial-time algorithm for this problem. All of these algorithms are based on the concepts of chain decomposable flows. A *chain flow* $\gamma = \langle w, P \rangle$ sends fixed units of flow w along a path P that obeys capacity constraints and length of the chain flow $\tau(\gamma)$ is less than the planning horizon T . $\tau(\gamma)$ equals the sum of the transit time of arcs in the path P . A feasible dynamic flow can be induced from any chain flow by sending w units of flow along P from time 0 until time $T - \tau(\gamma)$. A static flow can be decomposed into a set of chain flows and the corresponding dynamic flows induced by these chain flows are called chain decomposable flow. Their approach is not applicable to the maximum flow over time problem since the time-varying arc capacity makes the chain

decomposition of static flow invalid. Hoppe & Tardos (2000) also showed that there always exists an optimal flow over time that does not require the node storage except for sources and sinks, however, once the time-varying capacities are introduced, node capacity may need to be utilized in an optimal solution. Some DNFP problems with time-varying arc capacity have also been considered by several researchers (Ogier 1988, Fleischer 2001), but zero-transit times were assumed in their model. When all transit times are zero, the flow in an arc at any moment of time is independent of the flow in this arc in any previous moment. This independence allows for more flexible decomposition of the time horizon into time intervals.

In summary, although there is substantial body of research on DNFP, the time-varying capacity and non-zero transit time of maximum flow over time problem prevents the use of any existing fast algorithm.

2.4 Solution Methods for Maximum Flow over Time Model

We will adopt the uniform discretization approach to solve the maximum flow over time problem with piece wise constant arc capacity and constant finite node capacity. We discretize the continuous time into steps of unit length Δ . Denote τ_e and T as integral multiples of Δ , $\tau_e = m_e \cdot \Delta$, $m_e \in \{0, 1, 2, \dots\}$ and $T = p \cdot \Delta$. Construct a linear programming problem DP as follows.

$$\text{DP max } Q \equiv \sum_{k=0}^{p-1} \sum_{e \in \delta^-(S^-)} \hat{f}_e(k - m_e) \cdot \Delta$$

s.t.

$$\begin{aligned} & \Delta \left[\sum_{e \in \delta^-(v)} \hat{f}_e(k - m_e) - \sum_{e \in \delta^+(v)} \hat{f}_e(k) \right] \\ & = \hat{y}_v(k+1) - \hat{y}_v(k) \\ & \quad \forall v \in V \setminus S^+, k = 0, \dots, p-1 \\ & \hat{f}_e(k) \leq u_e(k\Delta) \quad k = 0, \dots, p-1; \forall e \in A \\ & \hat{y}_v(k) \leq z_v \quad k = 0, \dots, p; \forall v \in V \\ & \hat{f}_e(k) \geq 0 \quad k = 0, 1, \dots, p-1; \forall e \in A \\ & \hat{y}_v(k) \geq 0 \quad k = 0, 1, \dots, p; \forall v \in V \\ & \hat{f}_e(k) = 0; \hat{y}_v(k) = 0 \quad \forall k < 0; \forall e \in A; \forall v \in V \end{aligned}$$

$\hat{f}_e(k)$ and $\hat{y}_v(k)$ have the meaning of the flow rate at arc e and the node size at node v in the interval $[k\Delta, (k+1)\Delta]$ respectively. The objective of the problem is still to maximize the total flow to the sink in the discrete form. The constraints for flow balance and constraints for arc and node capacity are discretized. Fleischer & Tardos (1998) pointed out a strong connection between the continuous and discretized models. They show that if a continuous problem is feasible, there is a solution f that changes only at times in $\{1\Delta, 2\Delta, \dots, p\Delta\}$. So an optimal discrete-time solution can be transformed into an optimal continuous-time solution by sending flow at rate $\hat{f}(k)$ in the interval $[k\Delta, (k+1)\Delta]$. We call this *standard transformation procedure*. Thus, we can solve the maximum flow over time problem by solving linear programming problem DP, which can be solved by commercial mathematical programming solver like CPLEX.

In each iteration we need to calculate $s(g)$ which is a subgradient vector of average loss $g_N(\theta_k) = N^{-1} \sum_{j=1}^N L^j(\lambda^j, \theta_k)$, where θ_k is the vector of decision variable at iteration k , N is the number of replications in Monte Carlo sampling

and L denotes loss volume of a replication. If we know the subgradient vector for each replication $s(L)$, $s(g)$ can be calculate by averaging.

$$s(g) = N^{-1} \sum_{j=1}^N s(L^j)$$

2.5 Loss Subgradient Derivation

Loss volume in general was given in equation (1). In sample path analysis, the loss function L is the difference between total input stream and total flow to the sink Q . Q is the objective function value of problem DP. So L is

$$L = \sum_{i \in [1, \dots, |SIN|]} \int_0^T \lambda_i(t) dt - Q \quad (11)$$

The transformation procedure indicates the decision variable l , r , b of original network are represented as parameters u_e , z_v and m_e in problem DP. To achieve the analytical form, we ignore the small change of m_e due to small change of l , so that we can say the decision variables appear purely on the right hand side of constraints in problem DP. Only a subset of node capacities and arc capacities of network $\mathcal{N}$ corresponds to the decision variables. We denote them as $\{u_e : e \in A' \subset A\}$ and $\{z_v : v \in V' \subset V\}$. We first formulate the dual of problem DP as follows.

Dual of DP min

$$\sum_{e \in A} \sum_{k=0, \dots, p-1} u_e(k\Delta) \cdot w_e(k) + \sum_{v \in V} \sum_{k=0, \dots, p} z_v \cdot q_v(k)$$

s.t.

$$\Delta h_i(k + m_e) - \Delta h_j(k) + w_e(k) \geq 0$$

$$\forall e(i, j) \in A \setminus \delta^+(S^-); k = 0, \dots, p-1$$

$$\Delta h_i(k + m_e) - \Delta h_j(k) + w_e(k) \geq \Delta$$

$$\forall e(i, j) \in \delta^+(S^-); k = 0, \dots, p-1$$

$$h_v(k+1) - h_v(k) + q_v(k) \geq 0$$

$$\forall v \in V; k = 0, \dots, p$$

$$w_e(k) \geq 0$$

$$\forall e \in A; k = 0, \dots, p-1$$

$$q_v(k) \geq 0$$

$$\forall v \in V; k = 0, \dots, p$$

Let $\{u_e^0 : e \in A'\}$, $\{z_v^0 : v \in V'\}$ be arc and node capacities that correspond to the parameter values at current iteration and $\{u_e^*(k\Delta) : e \in A; k = 1, \dots, p-1\}$, $\{z_v^* : v \in V'\}$ the current right hand side value of problem DP. This implies $u_e^*(k\Delta) = u_e^0$ for $e \in A'$; $k = 1, \dots, p-1$ and $z_v^* = z_v^0$ for $v \in V'$. We use $\{w_e^*(k) : e \in A; k = 1, \dots, p-1\}$ and $\{q_v^*(k) : v \in V; k = 1, \dots, p\}$ to denote the optimal dual solution of problem DP corresponding to current right hand side value. We can obtain the loss subgradients with respect to decision variables from these dual variables. This is stated in the following theorem.

Theorem 1 The component of subgradient vector $s(L)$ corresponding to the parameter $u_e, e \in A'$ is $-\sum_{k=0}^{p-1} w_e^*(k)$; The component of subgradient vector $s(L)$ corresponding to the parameter $z_v : v \in V'$ is $-\sum_{k=0}^p q_v^*(k)$.

The Proof is in the Appendix B.

3. Numerical Example

We apply the above methodology to an example conveyor network parametric design. We consider a two-segment tandem system with a non-accumulation segment connected with an accumulation segment. Network representation is shown in Figure 4.

The network comprises three junctions. External arrival $\Lambda(t) = \mathcal{J}(t)$ occurs at junction 1. Λ is modeled as Markov Modulated Fluid

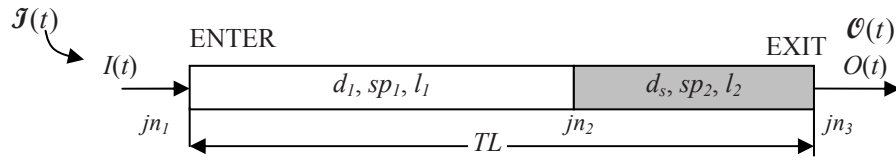


Figure 4 Two-segment Tandem conveyor system

Process (MMFP). Specifically the arrival rate at any given time is modulated by a continuous time Markov Chain with finite state space. In this example, four states that represent four arrival rates (including 0) are used. Junction 3 represents a sorter with capacity r_3 . The total length of two conveyor segments is TL . The decision variables are l_2 and r_3 , or $\theta = [l_2, r_3]^T$, since $l_1 = TL - l_2$.

For simplicity, we use linear cost function in this example. The total system cost consists of the cost of one non-accumulation conveyor, one accumulation conveyor and the sorter. Using practical values from the vendors, the total cost is $120l_1 + 300l_2 + 2500r_3$, and $l_1 = 700 - l_2$. We choose $P = 0.5\%$. Problem 1 becomes:

$$\begin{aligned} \text{Minimize } C(\theta) &= 180l_2 + 2500r_3 \\ \text{s.t. } E[\mathcal{L}(\Lambda, \theta)] &\leq E\left[\int_0^T \Lambda(t)dt\right] \times 0.5\% \\ l_2 &\leq 700. \end{aligned}$$

The corresponding approximate sample average Problem is now

$$\begin{aligned} \text{Minimize } C(\theta) &= 180l_2 + 2500r_3 \\ \text{s.t. } N^{-1} \sum_{j=1}^N L(\lambda^j, \theta) &\leq M \\ l_2 &\leq 700 \\ \lambda^1, \dots, \lambda^N &\text{ are random sample of } \Lambda. \\ M &= N^{-1} \left(\sum_{j=1}^N \int_0^T \lambda^j(t)dt \right) \times 0.5\%. \end{aligned}$$

We move the first constraint to the objective

function using penalty function

$$P(x) = \begin{cases} \frac{1}{2}x^2 & \text{if } x > 0 \\ 0 & \text{OW} \end{cases}$$

and penalty weight function $r_k = 1.2^k$. We

apply subgradient method to update the decision variables at each iteration, or $\theta_{k+1} = \theta_k +$

$$a_k \cdot \sigma \cdot s_k \text{ and } a_k = \frac{(U(\theta_k) - \bar{U}) \times 0.9^k}{\|\sigma \cdot s_k\|^2}, \text{ where}$$

$$U(\theta_k) = C(\theta_k) + r^k \cdot P\left(N^{-1} \sum_{j=1}^N L^j(\lambda^j, \theta_k) - M\right)$$

is the penalized objective function at iteration k ; $\bar{U}$ is an estimated lower bound on the total cost; constant row vector σ is a normalization factor used to adjust the big difference of magnitude of the two parameters. We choose $\sigma = [1, 1/10]$ in the example; s_k is the subgradient vector of U with respect to θ_k , it is calculated using the following formula.

$$s_k = \begin{bmatrix} 180 + r^k \text{Max}\left(N^{-1} \sum_{j=1}^N L^j(\lambda^j, \theta^k) - M, 0\right) \\ * N^{-1} \sum_{j=1}^N \frac{\partial L^j(\lambda^j, \theta^k)}{\partial l_2} \\ 2500 + r^k \text{Max}\left(N^{-1} \sum_{j=1}^N L^j(\lambda^j, \theta^k) - M, 0\right) \\ * N^{-1} \sum_{j=1}^N \frac{\partial L^j(\lambda^j, \theta^k)}{\partial r_3} \end{bmatrix}$$

The important calculations in the solution procedure are to evaluate the $L(\lambda, \theta)$ and its subgradient with respect to the decision variables for each replication in the sample and at each iteration. $L^j(\lambda^j, \theta_k)$ is evaluated through maximum flow over time model using (11). $\frac{\partial L^j(\lambda^j, \theta^k)}{\partial l_2}$ and $\frac{\partial L^j(\lambda^j, \theta^k)}{\partial r_3}$ are obtained using formulas stated in the Theorem 1.

Figure 5 shows the penalized objective value function changes over the iterations when $N = 5$.

Two curves represent the iterations starting from two different initial points; one is a feasible solution and another is infeasible in the beginning. They converge relatively fast to very close objective function values.

Sample path based optimization is implemented through the application of the deterministic programming algorithm for increasing value of N ; terminate the process when convergence of optimal objective function

value is conjectured. Many methods can be used for testing convergence ranging from simple graphic methods to formal statistical tests such as Geweke test (Geweke 1992). Figure 6 shows the quick convergence of solution for this example.

The uniform discretization algorithm to solve linear problem DP is a pseudo-polynomial algorithm since the running time depends polynomially on the planning horizon T . Problem DP has $p(n+m)$ non-negative variables and $p(2n+m)$ constraints, where $n:=|V|$, $m:=|A|$ and $p = T / \Delta$. Figure 7 shows the increase of computational time to solve DP using CPLEX as T increasing for a 10-segment model.

From running time magnitude on the graph, we can expect that current computers can handle the problem of networks with medium complexity over a time horizon of less than a few hours.

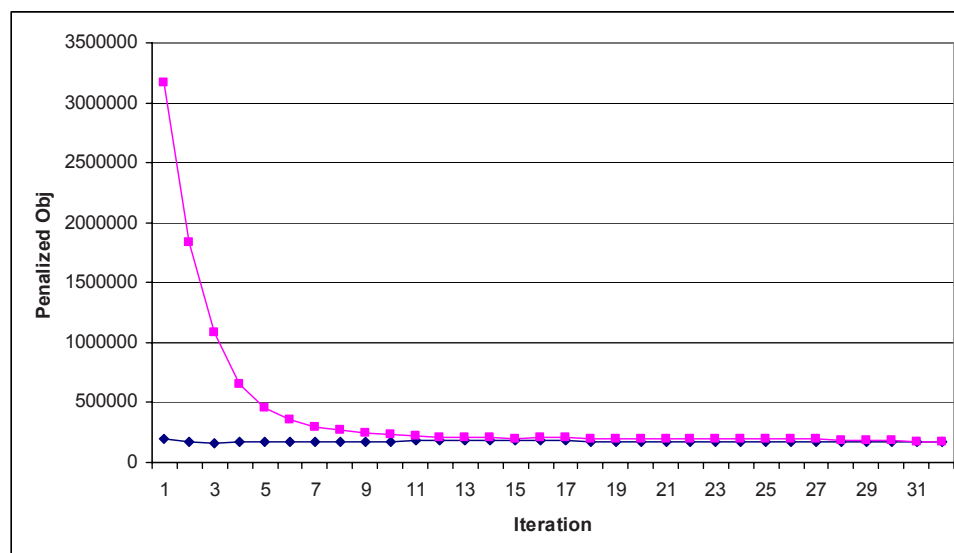


Figure 5 Penalized objective function when $N=5$

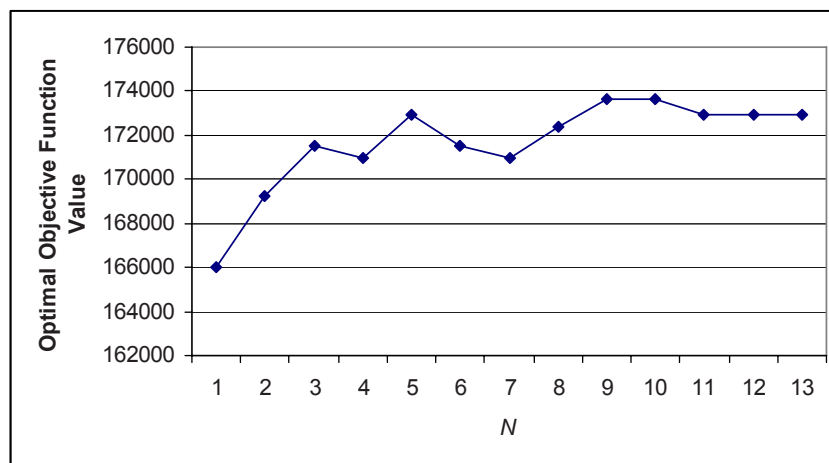


Figure 6 Convergence of optimal objective function value

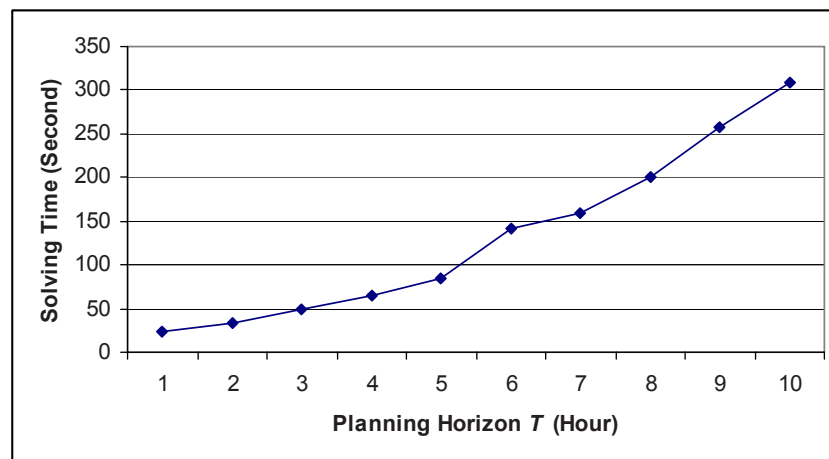


Figure 7 DNFP Solving time for a 10 segment network

4. Conclusions

In this research, we developed generic conveyor network models, identified several conveyor dynamic relationships. Based on these relationships, we used a Delay and Stock abstraction and we formulated a dynamic network flow model for the performance evaluation. We integrated this deterministic evaluation tool with the stochastic optimization techniques to search for the near optimal

conveyor network design with minimum cost and performance bounds. Several aspects of the analytical methods make analysis amenable in the designing of conveyor systems in distribution centers. First, the performance measure is a strictly convex function with respect to the design parameters. This allows the use of analytical tools that shorten the computation time tremendously. Secondly, the performance evaluation can be done through

commercial LP solver. Lastly, the exact subgradient information with respect to all parameters can be read out readily after solving the LP.

However, there are limitations of this approach. Delay and Stock Model is an approximate abstraction of the conveyor behavior. It ignores the spatial evolution of material flow on a conveyor. This leads to two possible error sources. First, the transfer time in Delay and Stock abstraction is assumed to be proportional to conveyor length. In reality, the blocking on accumulation conveyor depends on the accumulation position (the position at which the accumulation starts). Second, Delay and Stock Model always transmits the blocking from EXIT end of non-accumulation conveyor to ENTER end immediately, while many blockings are avoided through the material flow density change in real system. To show this effect, we conducted a series of two-segment experiments similar to that in the numerical example. We set $l_1 + l_2 = 150$. We fix the sorter speed but increase the accumulation position gradually. We collected the loss volume from the dynamic network flow model and loss volume from the fluid-based simulation approach (Wang & Zhou 2004). Figure 8 shows the comparison results.

We see that the Delay and Stock abstraction, and its dynamic network flow model, yields much higher loss volume when there is severe blocking. It yields lower loss volume when there is relatively large accumulation. This observation offers insights for further research to be addressed in the future research, along with the comparison to the use of the Batches Petri Net and fluid simulation model.

Appendices

Appendix A: Transformation Procedure

We describe the procedure to transform a problem that finds the loss volume in an A/M network $CN = (SM, JN, SIN, s)$ with a given input stream λ to an instance of maximum flow over time problem in three steps:

1) Select a base time unit Δ .

The choice of Δ should satisfy that all breakpoints of arc capacity functions in maximum flow over time problem and all arc transit times are integral multiples of Δ . The first condition means $\omega \geq \Delta$, where ω denotes the greatest common divisor of all breakpoints of λ functions. Generally, we consider the case when the capacities change over larger time scale than individual item movement. That means ω is relatively large. For the second condition we need to round up the real transit time l_k/v_k for all $k \in SM$ to the nearest multiple of Δ . The smaller Δ we choose, the less error there will be in the solution of maximum flow over time problem. However, a small value of Δ can also lead to prohibitive computational costs in solving the problem. Since this model will be used in the preliminary network design phase, a relatively larger Δ may be appropriate.

2) Determine the arc set and node set.

The node set consists of all ENTER and EXIT points of all segments and junctions plus an auxiliary ACC point for each accumulation type segment. The construction procedure is as follows:

a) For any $sm \in SM$ and $c(sm) = 'A'$, add an auxiliary position ACC between ENTER and EXIT of k .

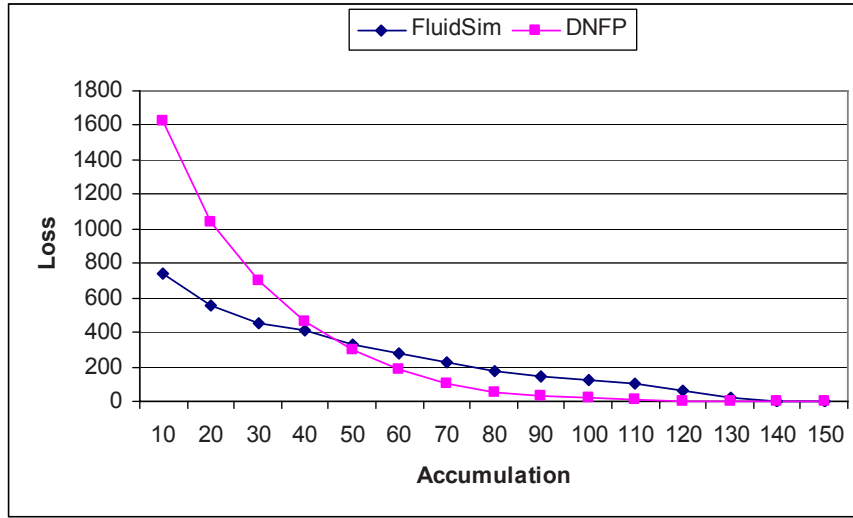


Figure 8 Loss volume comparison of FluidSim and DNFP

- b) $V = \text{ENTER}(k) \cup \text{ACC}(k) \cup \text{EXIT}(k)$ for all $k \in SM \cup JN$
- c) $e = (i, j) \in A$ if $\{\text{ENTER}(k) = i \text{ and } \text{EXIT}(k) = j \text{ for } k \in JN \text{ or } k \in SM \text{ and } c(k) = 'N'\}$ or $\{\text{ENTER}(k) = i \text{ and } \text{ACC}(k) = j \text{ for } k \in SM \text{ and } c(k) = 'A'\}$ or $\{\text{ACC}(k) = i \text{ and } \text{EXIT}(k) = j \text{ for } k \in SM \text{ and } c(k) = 'A'\}$
- 3) Determine the arc capacity, arc transit time and node capacity. The formulation is as follows:

a) $S^+ = \{\text{ENTER}(k) : k \in SIN\}; S^- = \text{EXIT}(s)$

b) $(i, j) \in A,$

$$\tau_{(i,j)} = \begin{cases} \lfloor l_k / v_k \rfloor & i = \text{ENTER}(k), j = \text{EXIT}(k), \\ & k \in SM, c(k) = 'N' \\ \lfloor l_k / v_k \rfloor & i = \text{ENTER}(k), j = \text{ACC}(k), \\ & k \in SM, c(k) = 'A' \\ 0 & i = \text{ACC}(k), j = \text{EXIT}(k), \\ & k \in SM, c(k) = 'A' \\ 0 & i = \text{ENTER}(k), j = \text{EXIT}(k), \\ & k \in JN \end{cases}$$

$$u_{(i,j)}(t) = \begin{cases} v_k d_k & i = \text{ENTER}(k), j = \text{EXIT}(k), \\ & k \in SM, c(k) = 'N' \\ v_k d_k & i = \text{ENTER}(k), j = \text{ACC}(k), \\ & k \in SM, c(k) = 'A' \\ v_k d_k & i = \text{ACC}(k), j = \text{EXIT}(k), \\ & k \in SM, c(k) = 'A' \\ r_k & i = \text{ENTER}(k), j = \text{EXIT}(k), \\ & k \in JN \setminus SIN \\ \lambda_k(t) & i = \text{ENTER}(k), j = \text{EXIT}(k), \\ & k \in SIN \end{cases}$$

c) $v \in V,$

$$z_v = \begin{cases} l_k d_k & v = \text{ACC}(k), \\ & k \in SM, c(k) = 'A' \\ b_k & v = \text{EXIT}(k), k \in JN \\ \infty & v = \text{ENTER}(k), k \in SIN, \\ & \text{or } v = \text{EXIT}(s) \\ 0 & \text{OW} \end{cases}$$

It can be verified that this transformation procedure enforced the solution of the maximum flow over time problem and satisfies all capacity and accumulation constraints in the original

network expressed through (2) to (8).

Appendix B: Proof for Theorem 1

Proof. We can write the objective value Q of problem DP as a function of right hand side value

$$Q(\{u_e(k\Delta) : e \in A, k = 0, \dots, p-1\}, \{z_v : v \in V\}).$$

From general LP theory, Q is a piece-wise linear and concave function of right hand side value.

According to strong duality theorem, we have

$$\begin{aligned} & \sum_{e \in A} \sum_{k=0, \dots, p-1} u_e^*(k\Delta) w_e^*(k) + \sum_{v \in V} \sum_{k=0, \dots, p} z_v^* \cdot q_v^*(k) \\ &= Q(\{u_e^*(k\Delta) : e \in A; k = 0, \dots, p-1\}, \\ & \quad \{z_v^* : v \in V\}) \quad (12) \end{aligned}$$

$\{w_e^*(k)\}$ and $\{q_v^*(k)\}$ are also a feasible solution to the dual problem corresponding to any $\{u_e(k\Delta)\}, \{z_v\}$, so according to the weak duality theory we have

$$\begin{aligned} & \sum_{e \in A} \sum_{k=0, \dots, p-1} u_e(k\Delta) w_e^*(k) + \sum_{v \in V} \sum_{k=0, \dots, p} z_v \cdot q_v^*(k) \\ & \geq Q(\{u_e(k\Delta) : e \in A; \forall k = 0, \dots, p-1\}, \\ & \quad \{z_v : v \in V\}) \quad (13) \end{aligned}$$

Subtract (12) from (13), we have

$$\begin{aligned} & \sum_{e \in A} \sum_{k=0, \dots, p-1} w_e^*(k)(u_e(k\Delta) - u_e^*(k\Delta)) \\ & + \sum_{v \in V} \sum_{k=0, \dots, p} q_v^*(k)(z_v - z_v^*) \geq \\ & Q(\{u_e(k\Delta) : e \in A; k = 0, \dots, p-1\}, \{z_v : v \in V\}) - \\ & Q(\{u_e^*(k\Delta) : e \in A; k = 0, \dots, p-1\}, \{z_v^* : v \in V\}) \quad (14) \end{aligned}$$

Since (14) is valid for any $\{u_e(k\Delta)\}, \{z_v\}$, we choose a particular one such that $u_e(k\Delta) = u_e^*(k\Delta), \forall e \in A \setminus A'; k = 0, \dots, p-1, z_v = z_v^*, \forall v \in V \setminus V'$ and $u_e(k\Delta) = u_e$ for

$e \in A'; k = 1, \dots, p-1$, (14) reduced to

$$\begin{aligned} & \sum_{e \in A'} \sum_{k=0, \dots, p-1} w_e^*(k)(u_e - u_e^*(k\Delta)) \\ & + \sum_{v \in V'} \sum_{k=0, \dots, p} q_v^*(k)(z_v - z_v^*) \geq \\ & Q(\{u_e : e \in A'; k = 0, \dots, p-1\}, \{z_v : v \in V'\}) - \\ & Q(\{u_e^*(k\Delta) : e \in A'; k = 0, \dots, p-1\}, \{z_v^* : v \in V'\}) \quad (15) \end{aligned}$$

Since $u_e^*(k\Delta) = u_e^0$ for $e \in A'; k = 1, \dots, p-1$ and $z_v^* = z_v^0$ for $v \in V'$, (15) further reduced to

$$\begin{aligned} & \sum_{e \in A'} \sum_{k=0, \dots, p-1} w_e^*(k)(u_e - u_e^0) \\ & + \sum_{v \in V'} \sum_k q_v^*(k)(z_v - z_v^0) \\ & \geq Q(\{u_e : e \in A'\}, \{z_v : v \in V'\}) \\ & - Q(\{u_e^0 : e \in A'\}, \{z_v^0 : v \in V'\}) \quad (16) \end{aligned}$$

That means $\left\{ \sum_{k=0}^{p-1} w_e^*(k) \right\}$ and

$\left\{ \sum_{k=0}^p q_v^*(k) \right\}$ are components of

subgradient of Q with respect to parameters. Theorem conclusion is reached because of the relationship between Q and L stated in equation (11).

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